

THADOMAL SHAHANI ENGINEERING COLLEGE

Department of Computer Engineering

Academic Year: 2025 – 2026

Semester: IV (January – May 2026)

Course Code: 2304212

Course Name: Basic Telecommunication Experiments

List of Experiments

1. Introduction to Cisco Packet Tracer networking simulation software tool and setting up a basic local area network (LAN) using switches and routers.
2. Identification and demonstration of modem, router, and switch hardware and understanding IP address.
3. Demonstration of signal transmission through wired media (twisted pair cable, coaxial and fibre optic cable).
4. Demonstration of signal transmission through wireless media.
5. Python program for generation and observation of amplitude modulated (AM) signals.
6. Python program for generation and observation of frequency modulated (FM) signals.
7. Python program for implementation and testing of error detection using Hamming code (11,7).
8. Python program for generation and observation of ASK, FSK, and PSK Signals.
9. Study and operation of a basic telephone system.
10. Study and operation of a basic mobile phone.
11. Case Study Presentation: Telecommunication Applications in Healthcare, Finance, E-Commerce or Education.

Subject In-Charge for MDM-FTS